

1. The term "push" and "pop" is related to the:  
A. Array                      B. Stacks  
C. Lists                      D. All of the above
2. Which of the following name does not relate to stacks?  
A. LIFO list                  B. Piles  
C. FIFO lists                D. Push-down lists
3. The time factor when determining the efficiency of algorithm is measured by:  
A. Counting microsecond  
B. Counting the number of statements  
C. Counting the number of key operations  
D. Counting the kilobytes of algorithm
4. Which of the following data structure store the homogeneous data elements?  
A. Array  
B. Pointers  
C. Records  
D. None of the above
5. When new data are to be inserted into a data structure, but there is no available space, this situation is usually called:  
A. Underflow                B. House full  
C. Overflow                  D. Saturated
6. The complexity of merge sort algorithm is:  
A.  $O(n \log n)$               B.  $O(n^2)$   
C.  $O(\log n)$                 D.  $O(2n \log n)$
7. Normally a computer cycle between:  
A. Fetch and halt states  
B. Fetch and execute status  
C. Fetch and interrupt states  
D. Fill and halt states.
8. Interrupts which are initiated by an instruction are:  
A. Internal                    B. Hardware  
C. External                   D. Software
9. The 8086 microprocessor has:  
A. 8 data lines                B. 14 data lines  
C. 12 data lines              D. 16 data lines
10. Which of the following registers is used to keep track of address of the memory location where the next instruction is located?  
A. Memory address register  
B. Instruction registers  
C. Memory data register  
D. Program Counter.
11. When a subroutine is called, the address of the instruction following the CALL instructions stored in/on the:  
A. Stack pointer              B. Program counter  
C. Accumulator               D. Stack
12. A subtractor is not usually present in a computer because:  
A. It is expensive  
B. It is not possible to design it  
C. The adder will take care of subtraction  
D. None of the above
13. The \_\_\_\_\_ circuit overcomes the problem of switching caused by jitter on the inputs.  
A. Astable multivibrator  
B. Bistable multivibrator  
C. Monostable multivibrator  
D. Schmitt trigger



14. Which of the following is not a form of multi-vibrator?  
A. Bistable                      B. Monostable  
C. Astable                      D. Tristable
15. The number of bits needed to encode all letters (26), 10 symbols and all numbers (10) is:  
A. 8                                  B. 6  
C. 7                                  D. 5
16. Which transistor element is used in CMOS logic?  
A. FET                              B. Bipolar  
C. MOSFET                      D. Unijunction
17. Three inputs NAND gate can be constructed using .....inputs NAND gates.  
A. 2                                  B. 4  
C. 3                                  D. 5
18. The initiation of a systems investigation may result from:  
A. A manager's format request  
B. An analyst's investigation  
C. A scheduled system's review  
D. All of the above
19. Which of the following need to be assessed during unit testing?  
A. Error handling  
B. Code stability  
C. Algorithmic performance  
D. None of above
20. Which of the following is a primary source of data?  
A. A student reads the voltage across a lamp using a voltmeter  
B. The meaning of a word is read from a dictionary  
C. A website provides information on the cost of air flights  
D. A sales brochure indicates what items are on special
21. Which of the following is used when a company goes outside its organization to develop a new system?  
A. Project dictionary  
B. System flowchart  
C. Request for proposal  
D. Data flow diagram
22. UML activity diagrams are useful in representing which analysis model elements?  
A. Behavioral elements  
B. Flow-based elements  
C. Class-based elements  
D. Scenario-based elements
23. The entity relationship diagram depicts:  
A. Relationship between the data objects  
B. Relationship between flow of data  
C. Input and output relationship  
D. Relationship between the activities and users
24. Which design model elements are used to depict a model of information represented from the user's views?  
A. Architectural design elements  
B. Data design elements  
C. Component-level design elements  
D. Interface design elements
25. In the relational modes, cardinality is termed as:  
A. Number of tuples  
B. Number of tables  
C. Number of attributes  
D. Number of constraints
26. The \_\_\_\_\_ function helps reduce the need to use multiple OR conditions.  
A. IN                                  B. AND  
C. Like                                  D. Between



27. The data-base environment does not have following component:  
 A. Users B. Data-base  
 C. Separate files D. Query language
28. Database management systems are intended to:  
 A. Eliminate data redundancy  
 B. Maintaining data integrity  
 C. Managing file access  
 D. All of the above
29. In "One to Many" cardinality one instance of a relation is mapped with:  
 A. One instance of second entity type  
 B. Primary key of 2 relation  
 C. Many instances of second entity type  
 D. None of above
30. The primary key is selected from the:  
 A. Composite keys  
 B. Candidate keys  
 C. Determinants  
 D. Foreign keys
31. The mechanism that brings a page into memory only when it is needed is called \_\_\_\_\_.  
 A. Segmentation B. Demand paging  
 C. Fragmentation D. Page replacement
32. Which of the following statements best explains a process?  
 A. It is a program  
 B. It is the program execution  
 C. It is the program that uses system calls  
 D. It is an instance of a program in execution
33. Virtual memory is \_\_\_\_\_.  
 A. An extremely large main memory  
 B. An extremely large secondary memory  
 C. An illusion of extremely large main memory  
 D. A type of memory used in super computers
34. The first thing that is searched when a command references a file is its:  
 A. I-node number  
 B. I-nodal number  
 C. I-node  
 D. Permission setting
35. Round robin scheduling is essentially the preemptive version of \_\_\_\_\_.  
 A. FIFO  
 B. Shortest remaining  
 C. Shortest job first  
 D. Longest time first
36. Let S and Q be two semaphores initialized to 1, where P0 and P1 processes the following statements.  
 Wait(S); wait (Q); ----; signal(S);  
 signal (Q) and wait (Q); wait(S); ----;  
 signal (Q); signal(S); respectively.  
 The above situation depicts a \_\_\_\_\_.  
 A. Semaphore B. Signal  
 C. Deadlock D. Interrupt
37. Context sensitive grammar (CSG) can be recognized by:  
 A. Finite state automata  
 B. Push down automata  
 C. 2-way linear bounded automata  
 D. None of these
38. In Lambda calculus computational model, a computation consists of an initial lambda expression plus a finite sequence of lambda terms, each deduced from the preceding term by one application of:  
 A. Alpha reduction  
 B. Beta reduction  
 C. Gamma reduction  
 D. Omega reduction



39. The set of all strings over the alphabet  $\Sigma = \{a, b\}$  (including  $\epsilon$ ) is denoted by:  
 A.  $(a+b)^*$  B.  $a+b^+$   
 C.  $(a+b)^+$  D.  $a^*b^*$
40. A language  $L$  is accepted by an FSA if it is:  
 A. CFL B. Recursive  
 C. CSL D. Regular
41. The objective of a semantic analysis is:  
 A. To break up the input source into small meaningful sequence of characters  
 B. Check if sequence of tokens forms a valid sequence  
 C. Check if syntactically correct statements make a meaningful reading  
 D. Walk through annotated parse tree and generate intermediate code
42. The final stage of a compiler design is:  
 A. Lexical analysis  
 B. Code optimization  
 C. Semantic analysis  
 D. Code generation
43. The purpose of syntax analysis (also known as parsing) is to:  
 A. Splits the input into tokens  
 B. Deleting tokens  
 C. Recombine split tokens  
 D. Checking token
44. What does a Lexical Analyzer do?  
 A. It produces an executable from a source file  
 B. It reports errors in the source code programs  
 C. It provides options to convert from one higher level language to another  
 D. It reads the input source program and produces as output a sequence of tokens
45. What are the classifications of a compiler?  
 A. Single-pass B. Load-and-go  
 C. Multi-pass D. All of the above
46. Assuming that one allows 256 depth values levels to be used, how many memory would  $512 \times 512$  pixel displays require to store the Z-Buffer?  
 A. 512K B. 1024K  
 C. 256K D. 128K
47. Computer memory organized as a two-dimensional array with each  $(x, y)$  addressable location corresponding to one pixel.  
 A. Pixel pointer B. Frame buffer  
 C. Array buffer D. Location arrays
48. When transforming a random Axis-Aligned Bounding Box define by the points  $(near_x, near_y, near_z)$  and  $(far_x, far_y, far_z)$  to the standard orthographic viewing box, which affine transforms are used?  
 A. Shear and translation  
 B. Scale and shear  
 C. Rotation and scale  
 D. Translation and scale
49. The amplifier circuit suitable for impedance matching is:  
 A. Common base  
 B. Common emitter  
 C. Common collector  
 D. None of the above
50. Which type of special-purpose diode has the characteristics of a voltage-controlled capacitor?  
 A. A tunnel diode B. A schottky diode  
 C. A Zener diode D. A varactor diode



51. If 60 J of energy are available for every 15 C of charge, what is the voltage?  
 A. 4V                      B. 15V  
 C. 60V                     D. 0.25V
52. Which of the following statement is not correct?  
 A. Voltage source is an active element  
 B. Resistance is a passive element  
 C. Current source is a passive element  
 D. Capacitance is a passive element
53. A device that changes one form of energy to another is called:  
 A. Rheostat              B. Transducer  
 C. Oscillator             D. Varicap
54. An ideal operational amplifier has:  
 A. Infinite output impedance  
 B. Infinite bandwidth  
 C. Zero input impedance  
 D. All of the above
55. What characteristics would characterize an ideal operational amplifier?  
 A. An infinite voltage gain, an infinite input resistance and zero output resistance  
 B. An infinite voltage gain, an infinite input resistance and an infinite output resistance  
 C. An infinite voltage gain, zero input resistance and an infinite output resistance  
 D. An infinite voltage gain, zero input resistance and zero output resistance
56. The BER is an abbreviation of:  
 A. Baud error rate    B. Bit excess rate  
 C. Bit error rate      D. Bit error ratio
57. Antennas commonly used for microwave links are:  
 A. Loop antennas  
 B. Paraboloidal dishes  
 C. Long-periodic antennas  
 D. Rombic antennas
58. The effect in non-linear detection of amplitude modulation where input signal is modulated by the noise is known as:  
 A. Noise insertion  
 B. Nonlinearity  
 C. Threshold effect  
 D. False alarm
59. In AM transmission, the frequency which is not transmitted is:  
 A. Audio Frequency  
 B. Lower side frequency  
 C. Upper side frequency  
 D. Carrier frequency
60. The maximum bandwidth is occupied by...  
 A. ASK                      B. BPSK  
 C. FSK                      D. None of above
61. Non-repudiation is an E-commerce security issue, this means:  
 A. To detect data manipulation by unauthorized parties  
 B. To assure content of information is available to authorized parties only  
 C. To prevent an entity from denying its previous commitments or action  
 D. To identify the two parties communicating with each other
62. Which is not the learning situation in the machine learning?  
 A. Rote learning  
 B. Learning by asking  
 C. Learning by example  
 D. Learning by analogy



63. Which generation of Web Technology is related to Semantic Web?  
 A. Web 4.0      B. Web 2.0  
 C. Web 3.0      D. Web 1.0
64. The DSA is an abbreviation of:  
 A. Digital signature algorithm  
 B. Desktop signature algorithm  
 C. Digital signature analysis  
 D. Desktop signature analysis
65. The data encryption standard (DES) and the advanced encryption standard (AES) are:  
 A. Block cipher  
 B. Loop cipher  
 C. Stream cipher  
 D. Quantum cipher
66. You are designing a network which needs to support 200 users. You don't plan to extend the segment beyond the current number of users. Which subnet mask would best meet your needs?  
 A. 255.255.0.0      B. 255.0.0.0  
 C. 255.255.255.0      D. 255.224.0.0
67. The RPC is an abbreviation of:  
 A. Remote Procedure Call  
 B. Remote Path Call  
 C. Remote Process Call  
 D. Remote Procedure Coverage
68. Which of the following is NOT a connection-oriented network?  
 A. X.25      B. Frame relay  
 C. ATM      D. UDP
69. The present IP address is defined by:  
 A. 32 bits      B. 64 bits  
 C. 48 bits      D. 96 bits
70. What is an example of a MAC address?  
 A. Az32:6362:2434  
 B. A625:cbdf:6525  
 C. BA:281x8288  
 D. 101:354:665:8734:ffd6:8023
71. Protocol Data Unit (PDU) of network layer is known as:  
 A. Packet      B. Datagram  
 C. Segment      D. Frame
72. Suppose there is an object of type person, which of the following can be considered as one of its attributes/s?  
 A. Name  
 B. Work ()  
 C. Age  
 D. Both name and age
73. Which of the following is not a valid relational operator?  
 A. ==      B. >=  
 C. =>      D. <=
74. Which of the following is not one of the processes that a high-level language program must go through before it is ready to executed?  
 A. Translation      B. Controlling  
 C. Loading      D. Linking
75. The minimum number of temporary variables needed to swap the contents of two variables is:  
 A. 0      B. 2  
 C. 1      D. 3
76. When following piece of code is executed, what happens?  
 b = 3;  
 a = b++;  
 A. A contains 3 and b contains 4  
 B. A contains 4 and b contains 3  
 C. A contains 4 and b contains 4  
 D. A contains 3 and b contains 3



77. The ASCII value of character "B" in hexadecimal is:

- A. 40                      B. 42  
C. 41                      D. 42

78. Which is not the part of an expert system?

- A. Working memory  
B. Knowledge base  
C. Inference engine  
D. Operating system

79. What is the term used for describing the judgmental or commonsense part of problem solving?

- A. Heuristic              B. Value based  
C. Critical                D. Analytical

80. Expert systems in AI is example of:

- A. Bottom-up approach  
B. Neuron network  
C. Top-down approach  
D. None of the above

## ANSWER SHEET

|      |      |      |      |      |      |      |      |      |      |
|------|------|------|------|------|------|------|------|------|------|
| 1.B  | 2.C  | 3.C  | 4.A  | 5.C  | 6.A  | 7.B  | 8.D  | 9.D  | 10.D |
| 11.D | 12.C | 13.D | 14.D | 15.B | 16.C | 17.A | 18.D | 19.A | 20.A |
| 21.C | 22.D | 23.A | 24.B | 25.A | 26.B | 27.C | 28.D | 29.C | 30.B |
| 31.B | 32.D | 33.C | 34.A | 35.A | 36.C | 37.C | 38.B | 39.A | 40.D |
| 41.C | 42.D | 43.C | 44.D | 45.D | 46.A | 47.B | 48.C | 49.C | 50.B |
| 51.A | 52.C | 53.B | 54.B | 55.A | 56.D | 57.B | 58.B | 59.A | 60.C |
| 61.C | 62.B | 63.C | 64.A | 65.A | 66.C | 67.A | 68.D | 69.A | 70.B |
| 71.A | 72.D | 73.C | 74.B | 75.A | 76.A | 77.B | 78.D | 79.B | 80.C |